

广东工业大学试卷参考答案及评分标准 (A)

课程名称: _____ 数字信号处理 _____

考试时间: 2024 年 7 月 2 日 (第 19 周 星期二)

一、 填空题 (每小题 2 分, 共 10 分)

1. $\frac{1-z^{-4}}{1-z^{-1}}, \frac{1-e^{-j2\pi k}}{1-e^{-j\frac{\pi}{2}k}}, k=0,1,2,3$

2. 不是

3. 12

4. $\frac{1}{1-0.5z^{-1}}, |z| > 0.5$

5. N

6. $N \log_2 N, 1, 2$

7. 一, $\frac{N-1}{2}$

8. 二, $\frac{N-1}{2}$

9. $\frac{1+3z^{-1}}{1+4z^{-1}}$

10. 不是

二、 计算题 (共 36 分)

1 (16 分)、解: (1) 由因果稳定性, 可知收敛域 ROC: $|z| > 0.2$

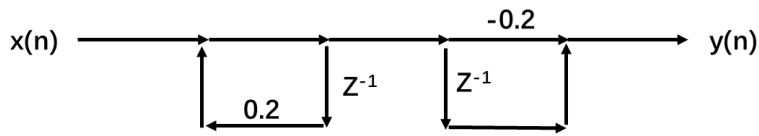
利用常用 z 变换对, $\frac{1}{1-0.2z^{-1}}, |z| > 0.2 \Leftrightarrow 0.2^n u(n)$, 因此,

$$\begin{aligned} H(z) &= \frac{z^{-1}-0.2}{1-0.2z^{-1}} = z^{-1} \frac{1}{1-0.2z^{-1}} - \frac{0.2}{1-0.2z^{-1}} \\ \Rightarrow h(n) &= 0.2^{n-1} u(n-1) - 0.2 \cdot 0.2^n u(n) \\ &= 0.2^{n-1} u(n-1) - 0.2^{n+1} u(n) \end{aligned}$$

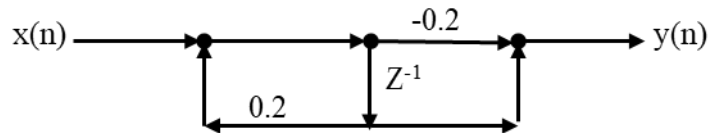
(2)

$$\begin{aligned} H(z) &= \frac{z^{-1}-0.2}{1-0.2z^{-1}} \Rightarrow \frac{Y(z)}{X(z)} = \frac{z^{-1}-0.2}{1-0.2z^{-1}} \Rightarrow y(n) - 0.2y(n-1) = x(n-1) - 0.2x(n) \\ \therefore y(n) &= -0.2x(n) + x(n-1) + 0.2y(n-1) \end{aligned}$$

(3) 直接 I 型结构信号流图



(4) 典范型结构信号流图



2 (10 分)、解：根据题意可知 $h(n) = \{4, 3, 2, 1\}$, $x(n) = \{1, 2, 3, 4\}$

(1) 计算线性卷积

$$\begin{array}{r}
 \begin{array}{cccc}
 4 & 3 & 2 & 1 \\
 1 & 2 & 3 & 4 \\
 \hline
 16 & 12 & 8 & 4 \\
 12 & 9 & 6 & 3 \\
 8 & 6 & 4 & 2 \\
 4 & 3 & 2 & 1 \\
 \hline
 4 & 11 & 20 & 30 & 20 & 11 & 4
 \end{array}
 \end{array}$$

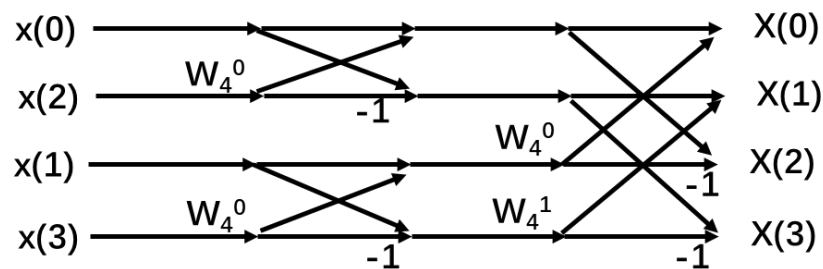
因此, $x(n)*h(n)=\{4, 11, 20, 30, 20, 11, 4\}$

(2) 计算圆周卷积

$$\begin{array}{r}
 \begin{array}{cccccc}
 4 & 11 & 20 & 30 & 20 & 11 & 4 \\
 20 & 11 & 4 & & & & \\
 \hline
 24 & 22 & 24 & 30 & & &
 \end{array}
 \end{array}$$

因此, 圆周卷积 $x(n) \textcircled{4} h(n) = \{24, 22, 24, 30\}$

3 (10 分)、



三、设计题 (共 34 分)

1 (16 分)、解: 确定数字频率: $\omega_p = 2\pi \cdot 3/15 = 0.4\pi, \omega_s = 2\pi \cdot 4.5/15 = 0.6\pi$

因此, 3dB 截止频率 $\omega_c = (\omega_p + \omega_s)/2 = 0.5\pi$, 过渡带 $\Delta\omega = \omega_s - \omega_p = 0.2\pi$

确定汉宁窗长度 N:

$$N = \Delta\omega / (\omega_s - \omega_p) = 6.2\pi / 0.2\pi = 31$$

群延时 $\tau = (N-1)/2 = 15$

$$\text{此时, } h_d(n) = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)} = \frac{\sin(0.5\pi(n-15))}{\pi(n-15)}$$

当 $n=15$ 时, 不加窗, 取 $h(15) = \frac{\omega_c}{\pi} = \frac{0.5\pi}{\pi} = 0.5$

当 $n \neq 15$ 时, 加窗, 取

$$h(n) = h_d(n)w(n) = \frac{\sin(0.5\pi(n-15))(1-\cos(\pi n/15))R_{30}(n)}{2\pi(n-15)}$$

$$\text{因此, } h(n) = \begin{cases} \frac{\sin(0.5\pi(n-15))(1-\cos(\pi n/15))R_{30}(n)}{2\pi(n-15)}, & n \neq 15 \\ 0.5, & n = 15 \end{cases}$$

2 (18 分)

解: 确定数字指标: $T=1/f_s=0.5 \cdot 10^{-5}$ 秒

$$\omega_p = 2\pi \frac{10}{100} = 0.2\pi, \delta_p = 2dB$$

$$\omega_s = 2\pi \frac{20}{100} = 0.4\pi, \delta_s = 15dB$$

考虑预畸变, 确定模拟指标:

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right) = 2 \cdot 100K \cdot \tan(0.1\pi) = 0.6429 \cdot 10^5, \delta_p = 2dB$$

$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) = 2 \cdot 100K \cdot \tan(0.2\pi) = 1.4531 \cdot 10^5, \delta_s = 15dB$$

确定 Butterworth 模拟低通滤波器参数

计算阶数 N:
$$N = \left\lceil \frac{\lg\left(\frac{(10^{0.2} - 1)}{(10^{1.5} - 1)}\right)}{2 \cdot \lg(0.6429/1.4531)} \right\rceil = \lceil 2.4269 \rceil = 3$$

计算 3dB 截止频率 $\Omega_c = 0.6429 \cdot 10^5 \cdot (10^{0.2} - 1)^{-1/6} = 7.0301 \cdot 10^4$

查表，归一化系统函数:
$$H_{aN}(s) = \frac{1}{s^3 + 2s^2 + 2s + 1}$$

去归一化，模拟滤波器系统函数:

$$H_a(s) = H_{aN}\left(\frac{s}{\Omega_c}\right) = \frac{1}{\left(\frac{s}{\Omega_c}\right)^3 + 2\left(\frac{s}{\Omega_c}\right)^2 + 2\left(\frac{s}{\Omega_c}\right) + 1}$$

根据双线性变换的映射公式，确定 Butterworth 数字低通滤波器的系统函数:

$$H(z) = H_a(s) \Big|_{s=\frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}} = \frac{1}{\left(\frac{2}{\Omega_c T} \frac{1-z^{-1}}{1+z^{-1}}\right)^3 + 2\left(\frac{2}{\Omega_c T} \frac{1-z^{-1}}{1+z^{-1}}\right)^2 + 2\left(\frac{2}{\Omega_c T} \frac{1-z^{-1}}{1+z^{-1}}\right) + 1}$$